

Agent Swarm Review Report

Sheet Metal Cell Phone Stand -- POP Writeup

CTE Metals, Bend-La Pine Schools

Instructor: Andrew McAteer

14

Students Reviewed

108

Clarifying Questions

3

Criteria Complete

23

Criteria Missing

Generated: February 23, 2026 at 06:07 PM

Submission Overview

Student	Course	Quality	Design	Procedure	Materials	Reflection
Student 11921	P3 Metals 1	PARTIAL	~	~	~	X
Student 21134	P3 Metals 1	PARTIAL	~	~	~	~
Student 8384	P3 Metals 1	PARTIAL	X	~	~	!
Student 4671	P3 Metals 2	PARTIAL	~	~	~	OK
Student 8370	P3 Metals 2	MINIMAL	X	~	~	X
Student 11596	P5 Metals 1	PARTIAL	!	~	~	OK
Student 1320	P5 Metals 1	PARTIAL	~	~	~	~
Student 1768	P5 Metals 1	PARTIAL	~	~	~	X
Student 20623	P5 Metals 1	PARTIAL	~	~	X	!
Student 20756	P5 Metals 1	MINIMAL	X	X	X	X
Student 8348	P5 Metals 1	PARTIAL	~	~	~	OK
Student 8430	P5 Metals 1	PARTIAL	X	~	~	X
Student 10263	P5 Metals 2	MINIMAL	X	X	X	X
Student 4823	P5 Metals 2	MINIMAL	X	X	X	X



Complete



Partial



Missing/Minimal

Common Patterns Across Submissions

Hot glue on metal: Flagged on 5+ students. Students choosing hot glue for sheet metal assembly instead of welding, riveting, or mechanical fasteners.

Missing reflection sections: Most students skipped or left blank the reflection component.

Template text left in: Several students submitted documents with template instructions, highlighted placeholders, or 'Wood Required' prompts still visible.

Missing BOM/tools lists: Even students with good procedures failed to provide a consolidated materials and tools list.

Vague procedure steps: Many procedures lack specific dimensions, bend angles, and tool callouts.

P3 Metals 1

Student 11921

[PARTIAL] Plan of Procedure - Sheet Metal Phone Dock.pdf

[PARTIAL] Design Intent & Problem Statement

States objective to design a functional phone stand. Lists skills practiced. No explicit problem statement or justification for material/angle choices.

- ? Could you expand on what specific problem or need your design aims to solve for someone using a phone stand?
- ? You chose 20-22 gauge galvanized steel. What were your reasons for selecting this specific type and thickness of metal?

[PARTIAL] Plan of Procedure (Step-by-Step)

Numbered procedure broken into phases. Includes some tools and safety reminders. Lacks specific measurements or angles for bends.

- ? Could you add specific measurements for the total length, width, and distances for each bend line on your flat pattern?
- ? What specific angles are you planning for 'The Cradle' and 'The Back/Stand' bends?

[PARTIAL] Materials & Tools List

Lists material type and various tools. Specific dimensions for the material and quantity are not provided.

- ? Could you add the specific dimensions (length and width) and quantity needed for one stand?
- ? Are there any other tools for measuring angles or final finishing that should be added?

[MISSING] Reflection & Writing Quality

Well-organized writing. 'Quality Check' section is a rubric, not a personal reflection.

- ? Could you add a section reflecting on your experience -- most challenging part, what you learned, what you'd do differently?

Flags:

- Missing specific dimensions/quantity
- Missing angles in procedure
- Quality Check is a rubric, not reflection
- Placeholder images for finish product

Student 21134

[PARTIAL] pop metal phone stand.docx

[PARTIAL] Design Intent & Problem Statement

Provides design concept name ('The Simple Slant') and specs. No explicit problem statement or material justification.

- ? What specific problem does 'The Simple Slant' solve for a cell phone user?
- ? What are the advantages of 22-gauge galvanized steel for a cell phone stand?

[PARTIAL] Plan of Procedure (Step-by-Step)

Phases with tools and measurements. Unclear how 'Back Supports' and 'Under support' are created. Hot glue vs. weld gap.

- ? How are the 'Back Supports' and 'Under support' created from your 3" x 7" flat pattern?
- ? Which assembly method do you plan to use -- hot glue or welding -- and what safety precautions are needed?

[PARTIAL] Materials & Tools List

Identifies material and tools throughout but no consolidated list.

- ? Could you provide a single, consolidated list of all materials with quantities and dimensions?
- ? Please create a complete list of all tools from layout to final assembly.

[PARTIAL] Reflection & Writing Quality

Clear writing. 'Design Ideas to Enhance' shows thought. No dedicated reflection on planning process.

- ? Could you reflect on the planning process? What was the most challenging part?

Flags:

- No explicit problem statement
- Materials/tools not consolidated
- Hot glue vs. weld ambiguity
- No dedicated reflection

Student 8384

[PARTIAL] Iain Bailes Phone Holder POP.pdf

[MISSING] Design Intent & Problem Statement

No design intent or problem statement section. Title implies a 'Phone Holder' but nothing explicit.

- ? What specific problem were you trying to solve with this phone holder design?
- ? Why did you choose sheet metal, and how did that influence your design?

[PARTIAL] Plan of Procedure (Step-by-Step)

4 numbered steps starting with cardboard prototype. 'Sheet metal cutter' and 'Plasma CNC' mentioned. Hot glue for assembly.

- ? Could you specify which type of sheet metal cutter you plan to use and why?
- ? What alternative metal joining techniques did you consider instead of hot glue?

[PARTIAL] Materials & Tools List

BOM table provided with dimensions and calculations. No separate tools list.

- ? Could you add a separate section listing all tools and safety equipment?
- ? Since you mentioned hot glue, could you add adhesive to your BOM?

[MINIMAL] Reflection & Writing Quality

No reflection section. Template instructions (yellow/orange highlights) still present. Pages 4-7 blank.

- ? Could you add a reflection on your design and planning process?
- ? Please remove template instructions and highlighted text.

Flags:

- Missing design intent
- Missing tools list
- Template instructions not removed
- Blank pages 4-7
- No dimensioned drawing

P3 Metals 2

Student 4671

[PARTIAL] Sheet Metal P.O.P - AG.pdf

[PARTIAL] Design Intent & Problem Statement

Identifies problems: lack of horizontal stands and durability. No justification for material/angle choices.

- ? How does 22-gauge galvanized steel specifically make your stand more durable?
- ? How do your chosen angles ensure the phone is held securely horizontally?

[PARTIAL] Plan of Procedure (Step-by-Step)

Lists general steps. Lacks specific dimensions, detailed tool callouts. Bending not mentioned.

- ? Could you add specific dimensions for the shapes you drew and cut?
- ? Can you describe where and how you planned to bend the metal?

[PARTIAL] Materials & Tools List

Lists steel and hot glue. Missing dimensions, quantity, and comprehensive tools list.

- ? Could you specify dimensions and quantity for the galvanized steel?
- ? Could you create a separate list of all tools used?

[COMPLETE] Reflection & Writing Quality

Thoughtful reflection on creativity, hot glue safety incident, and material durability. Clear and professional.

- ? What specific aspects allowed you to be most creative?
- ? What observations led you to conclude hot glue was durable?

Flags:

- Missing dimensions in BOM
- No dedicated tools list
- Photos section empty

Student 8370

[MINIMAL] Dawson phone holder POP.docx

[MISSING] Design Intent & Problem Statement

Jumps directly into procedural steps. No problem statement or design justification.

- ? What specific problem were you trying to solve with this stand?
- ? How do the dimensions help your stand meet a specific user need?

[PARTIAL] Plan of Procedure (Step-by-Step)

Numbered list with some dimensions. Lacks tool callouts. Deburring after gluing seems out of order. Hot glue for assembly.

- ? What specific tools would you use for cutting safely?
- ? What more permanent methods for joining sheet metal could you consider?

[PARTIAL] Materials & Tools List

Identifies starting sheet size. Missing metal type/gauge and comprehensive tools list.

- ? What specific type and gauge of sheet metal did you plan to use?
- ? Could you list all tools needed for each step?

[MISSING] Reflection & Writing Quality

No reflective component. Spelling error: 'Coutout'.

- ? What did you learn while planning this stand?
- ? What would you change or improve if making it again?

Flags:

- Missing 3 of 4 sections
- Hot glue for metal assembly
- Spelling error: 'Coutout'

P5 Metals 1

Student 11596

[PARTIAL] Copy of POP Template.pdf

[MINIMAL] Design Intent & Problem Statement

Title implies purpose. Reflection mentions triangle cut for charger. No explicit problem statement.

- ? What specific problem were you trying to solve for an iPhone user?
- ? How does the triangle cut for the charger address a specific need?

[PARTIAL] Plan of Procedure (Step-by-Step)

13 numbered steps with dimensions and some tools (scribe, tin snips, center punch, rivets). Missing bending tool details and safety callouts.

- ? What specific tools did you use for 90-degree bends?
- ? What is a 'triangle punch' and why did you choose hot glue over other joining methods?

[PARTIAL] Materials & Tools List

BOM table lists two sheet metal pieces. No consolidated tools list.

- ? Can you specify the exact type of sheet metal (mild steel, aluminum)?
- ? Could you create a consolidated list of all tools?

[COMPLETE] Reflection & Writing Quality

Discusses benefit of drawing steps, success of charger opening, and base stability improvement. Clear and professional.

- ? How did drawing every step help you make the process more efficient?
- ? What specific design changes would improve base stability?

Flags:

- No explicit problem statement
- Missing consolidated tools list
- Missing 3-view drawing
- Template placeholder for teacher initials

Student 1320

[PARTIAL] Morris PhoneStand POP.docx

[PARTIAL] Design Intent & Problem Statement

Describes 'Triple-Stack Slot' design for iPhone 14. Material thickness and pocket depth justified.

- ? What specific problem does your 'Triple-Stack Slot' stand solve?
- ? Why was 0.5-inch thick sheet metal chosen for a laminated assembly?

[PARTIAL] Plan of Procedure (Step-by-Step)

Assembly steps 2-4 detailed. Step 1 missing. Super glue and fingers for clamping. 75-degree tilt specified.

- ? Could you add Step 1 describing how individual components are prepared?
- ? What specific tools and measurements ensure parts fit correctly?

[PARTIAL] Materials & Tools List

Lists sheet metal and super glue. Component List section empty.

- ? Could you provide a complete list of all individual metal components with dimensions?
- ? What tools are needed to cut, measure, and prepare the metal pieces?

[PARTIAL] Reflection & Writing Quality

Clear writing with good technical terms. Safety note included. No explicit reflection section.

- ? What did you learn planning the 'Triple-Stack Slot' design?
- ? What would you change about your design or procedure?

Flags:

- Missing Step 1
- Empty Component List
- Title says 'sheet box' not 'cell phone stand'
- No reflection section

Student 1768

[PARTIAL] Sheet Metal Phone Stand POP.docx

[PARTIAL] Design Intent & Problem Statement

States project purpose. Mentions challenge of cutting hands. Contradiction: hot glue vs. folding metal.

? Can you clarify: hot glue or folding? Why did you change your approach?

? What features allow your stand to hold a phone both vertically and horizontally?

[PARTIAL] Plan of Procedure (Step-by-Step)

Phased procedure (1.1-1.5). General actions without specific measurements or tool callouts.

? Could you add specific dimensions and cutting tools for each part?

? What measurements or angles ensure the phone rest holds securely?

[PARTIAL] Materials & Tools List

Lists 'scrap sheet metal' without gauge. Tools categorized. 'Hope puncher' typo.

? Could you specify thickness and dimensions of the sheet metal?

? What is a 'Hope puncher'? (likely 'hole puncher')

[MISSING] Reflection & Writing Quality

Reflection section completely blank. Informal language ('damn hands', 'My brain'). PPE lists 'Common Sense' instead of physical gear.

? Remember to fill in the PROJECT REFLECTION after building your stand.

? How might you rephrase informal language for professional documentation?

Flags:

- Reflection blank
- Hot glue vs. folding contradiction
- 'Hope puncher' typo
- 'Common Sense' listed as PPE

Student 20623

[PARTIAL] Phone Holder POP.docx

[PARTIAL] Design Intent & Problem Statement

Clear goal: 'functional desktop art' with iPhone orientation and MagSafe compatibility. Triangular aesthetic identified.

? What properties of galvanized sheet metal make it good for 'functional desktop art'?

? Were there structural reasons for the triangular aesthetic beyond visual impact?

[PARTIAL] Plan of Procedure (Step-by-Step)

Logical phases (Prototyping, CAD, Cutting, Cold-Forming, Integration). Key tools mentioned. Missing measurements and safety.

? Could you add specific measurements for mock-up dimensions and bend angles?

? What safety precautions for ArcLight Plasma CNC and angle grinder?

[MISSING] Materials & Tools List

'Bill of Materials & Tooling' section present but completely empty.

? Could you fill in the BOM with material dimensions, quantity, and all tools?

[MINIMAL] Reflection & Writing Quality

Well-organized, professional writing. 'Final Evaluation' checks product but lacks process reflection.

? Could you add a reflection on the fabrication process -- challenges, learning?

Flags:

- BOM section completely empty
- Final Evaluation is checklist, not reflection

Student 20756

[MINIMAL] Pop template - Hans Phone stand.docx

[MISSING] Design Intent & Problem Statement

Contains template instructions only. No student content for design intent.

- ? What specific problem is your stand designed to solve?
- ? What were your initial design ideas?

[MISSING] Plan of Procedure (Step-by-Step)

'PLAN STEPS:' heading present. No actual steps from student.

- ? Could you walk through the main steps from cutting to finishing?
- ? What tools and safety precautions for each step?

[MISSING] Materials & Tools List

BOM heading present. Prompt says 'Wood Required' for a metal project.

- ? Could you list all materials including type, dimensions, and quantity?
- ? What tools were needed for each fabrication step?

[MISSING] Reflection & Writing Quality

No student-written reflection. Only template text.

- ? What was the most challenging part?
- ? What would you do differently?

Flags:

- Mostly template -- very little student content
- 'Wood Required' on a metal project
- All 4 criteria missing

Student 8348

[PARTIAL] Inman Phonestand POP.pdf

[PARTIAL] Design Intent & Problem Statement

Clearly states problem. Aims for 'elegant' design minimizing material. Target phone specified.

- ? How did wanting elegance and minimal waste influence specific design choices?
- ? Beyond holding a phone, what properties of sheet metal made it a good choice?

[PARTIAL] Plan of Procedure (Step-by-Step)

Iterative process documented. Mentions sheet metal cutter and cornice brake. Documented what actually happened, not what was planned.

- ? Could you add dimensions and angles for your final prototype?
- ? Which type of cutter did you use and why?

[PARTIAL] Materials & Tools List

BOM lists initial scrap (18 gauge, 12x12 galvanized). Missing final dimensions and consolidated tools list.

- ? Could you add the final dimensions for your finished stand?
- ? Could you create a separate tools list?

[COMPLETE] Reflection & Writing Quality

Thoughtful reflection on cardboard vs. sheet metal properties. Clear articulation of iterative progression.

- ? How did understanding material differences change your final design?
- ? What was most surprising about moving from cardboard to metal?

Flags:

- Lost dimensioned sketch (acknowledged)
- Teacher initials column empty
- No consolidated tools list

Student 8430

[PARTIAL] phone holder POP Template.pdf

[MISSING] Design Intent & Problem Statement

No problem statement or design intent section. Purpose implied by title only.

- ? What problem or need were you addressing?
- ? Why did you choose these specific dimensions and triangular shape?

[PARTIAL] Plan of Procedure (Step-by-Step)

12 numbered steps with some tools and measurements. Relies heavily on hot glue. Some unclear step details.

- ? What safety precautions for collecting and cutting sheet metal?
- ? Is the 1/2x3" piece separate or a bent section of the 6.5" piece?

[PARTIAL] Materials & Tools List

BOM table provided. Thickness column blank. No separate tools list. Dimensions differ from procedure.

- ? Please fill in the thickness for each piece.
- ? How do BOM dimensions relate to procedure cut dimensions?

[MISSING] Reflection & Writing Quality

No dedicated reflection section.

- ? What did you learn while developing this plan?
- ? What changes might you make to your design?

Flags:

- Missing design intent
- Missing reflection
- BOM vs. procedure dimension discrepancy
- No dimensioned drawing

P5 Metals 2

Student 10263

[MINIMAL] POP Template.docx

[MISSING] Design Intent & Problem Statement

Contains drawing prompt only. No student content.

- ? What specific problem is your stand designed to solve?
- ? Why did you choose certain materials or angles?

[MISSING] Plan of Procedure (Step-by-Step)

'PLAN STEPS:' heading and instruction only. No steps from student.

- ? Could you list specific numbered steps to fabricate your stand?
- ? What tools, measurements, and safety for each step?

[MISSING] Materials & Tools List

BOM heading with 'Wood Required' -- wrong material for sheet metal project.

- ? What type and dimensions of sheet metal will you use?
- ? What tools do you anticipate needing?

[MISSING] Reflection & Writing Quality

Template prompts only. No student reflection.

- ? What did you find most challenging during planning?
- ? What would you approach differently?

Flags:

- Template only -- no student content
- 'Wood Required' for metal project
- All 4 criteria missing

Student 4823

[MINIMAL] 1771870447.80859.jpg (photo only)

[MISSING] Design Intent & Problem Statement

Photo shows completed stand. No written problem statement or design justification.

- ? What specific problem were you solving with this stand?
- ? Why did you choose this type of metal?

[MISSING] Plan of Procedure (Step-by-Step)

Photo shows finished product with cuts and bends. No written procedure.

- ? Could you outline the step-by-step process from start to finish?
- ? What tools did you use for cutting and bending?

[MISSING] Materials & Tools List

Photo shows sheet metal material. No written list.

- ? What kind of sheet metal and starting dimensions?
- ? Could you list all tools used?

[MISSING] Reflection & Writing Quality

Photo only. No written reflection. Stand shows raw finish with surface scratches and tool marks.

- ? What was the most challenging part of making your stand?
- ? What would you do differently next time?

Flags:

- Photo only -- entire writeup component missing
- Raw metal finish with scratches
- No written content at all